

Ivanna Aleman-Coronado

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EDUCATION

The University of Texas at Austin – Austin, TX

May 2028

B.S. Computer Science, Minor in Robotics

- Relevant Coursework: Computer Architecture, Autonomous Driving, Data Structures, Discrete Math for Computer Science, Calculus II & III,, Robotics.
- Robotics Honors Program, *Freshman Research Initiative Participant*

Carroll High School – Southlake, TX

May 2025

PROFESSIONAL EXPERIENCE

Undergraduate Student Researcher, University of Texas at Austin (UT CS Autonomous Driving Lab)

Jan 2026 - Present

Austin, Texas

- Participating in hands-on research on autonomous systems, including perception, planning, and control for scaled autonomous vehicles.
- Implementing multi-sensor fusion using camera, LiDAR, and IMU data to detect and track car environments in real time.
- Developing algorithms for long-term localization, mapping, and decision-making in dynamic scaled race track environments with plans to move to a dynamic multi-agent space.

Machine Learning Research Intern, University of Texas at Dallas (StraRLiNG Lab)

May 2024 - August 2024

Dallas, Texas

- Developed anomaly-based machine learning models using K-means Clustering, Gaussian Mixture Models, and One-Class classifiers to develop solutions for real-world Denial-of-service attacks.
- Conducted comparative analysis of signature-based versus anomaly-based detection methods on network traffic data, evaluating performance across accuracy, recall, and precision metrics.
- Collaborated with a research team to build a detection system that outperformed traditional signature-based approaches for identifying novel attack patterns.
- Presented findings on machine learning-based network security to UTD faculty.

PROJECTS

Block-Based Programming Game (Squirrel++) | Java, Java Swing | GIMP 3 (level Design)

- Developed an interactive educational game featuring a drag-and-drop visual programming interface that allows players to construct logic-driven programs to control a character in a 2D environment.
- Engineered a custom command system with modular components including movement commands, conditional logic (IF/ELSE statements), and game state conditions, executed through a runtime program interpreter.
- Designed an intuitive UI using Java Swing with layered panels, custom rendering, and dynamically editable conditional blocks that update behavior in real-time.
- Implemented collision detection, state-based movement, and level data evaluation to enable responsive gameplay and real-time decision-making
- Created an educational tool prototype for elementary coding classes, demonstrating proficiency in object-oriented design and event-driven programming.

Personal Portfolio Website | JavaScript, HTML, CSS | Figma (UI Design)

- Developed a responsive personal portfolio website from scratch to showcase professional experience, technical projects, and skills.
- Implemented interactive UI components and dynamic content rendering using vanilla JavaScript to enhance user engagement.
- Utilized CSS adaptive layouts to ensure cross-browser compatibility and optimal viewing across desktop, tablet, and mobile devices.
- Won first place in Hispanic Association of Computer Scientists Website Competition, selected from submission by UT Austin CS Students.

AWARDS AND RECOGNITIONS

- **HACS Personal Website Competition winner:** Awarded first place for designing and developing a personal portfolio website, demonstrating technical proficiency and web design.
- **Hispanic Scholarship Fund Scholar:** Selected from 70,000 applications for academic achievement and leadership.
- **National Hispanic Recognition Award:** Honored by the College Board for outstanding achievement among Latino and Hispanic high school students.

TECHNICAL SKILLS

Languages: Java, Python, C, Javascript, Swift, HTML/CSS, React

Technologies: Linux, Version Control (Git), VS Code PyCharm, JFrame/JPanel, FoxGlove, ROS navigation stack.

Work Eligibility: Authorized to work in the U.S.